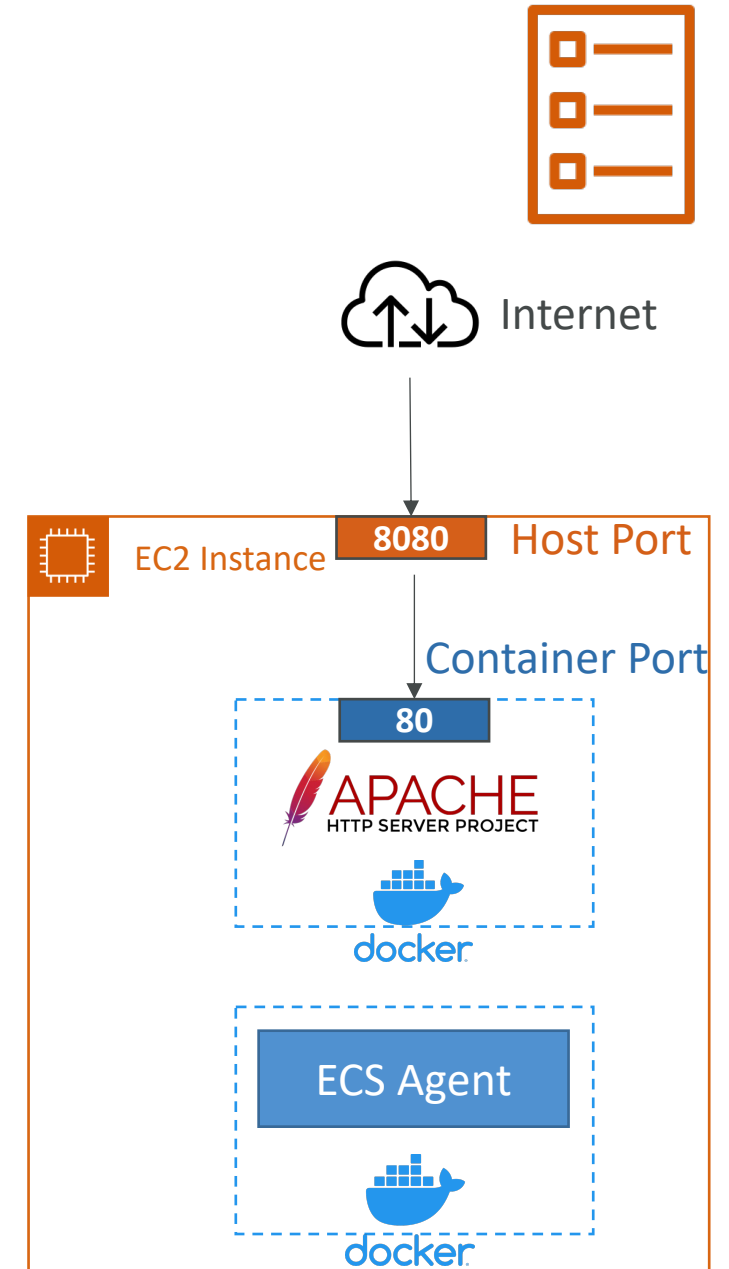


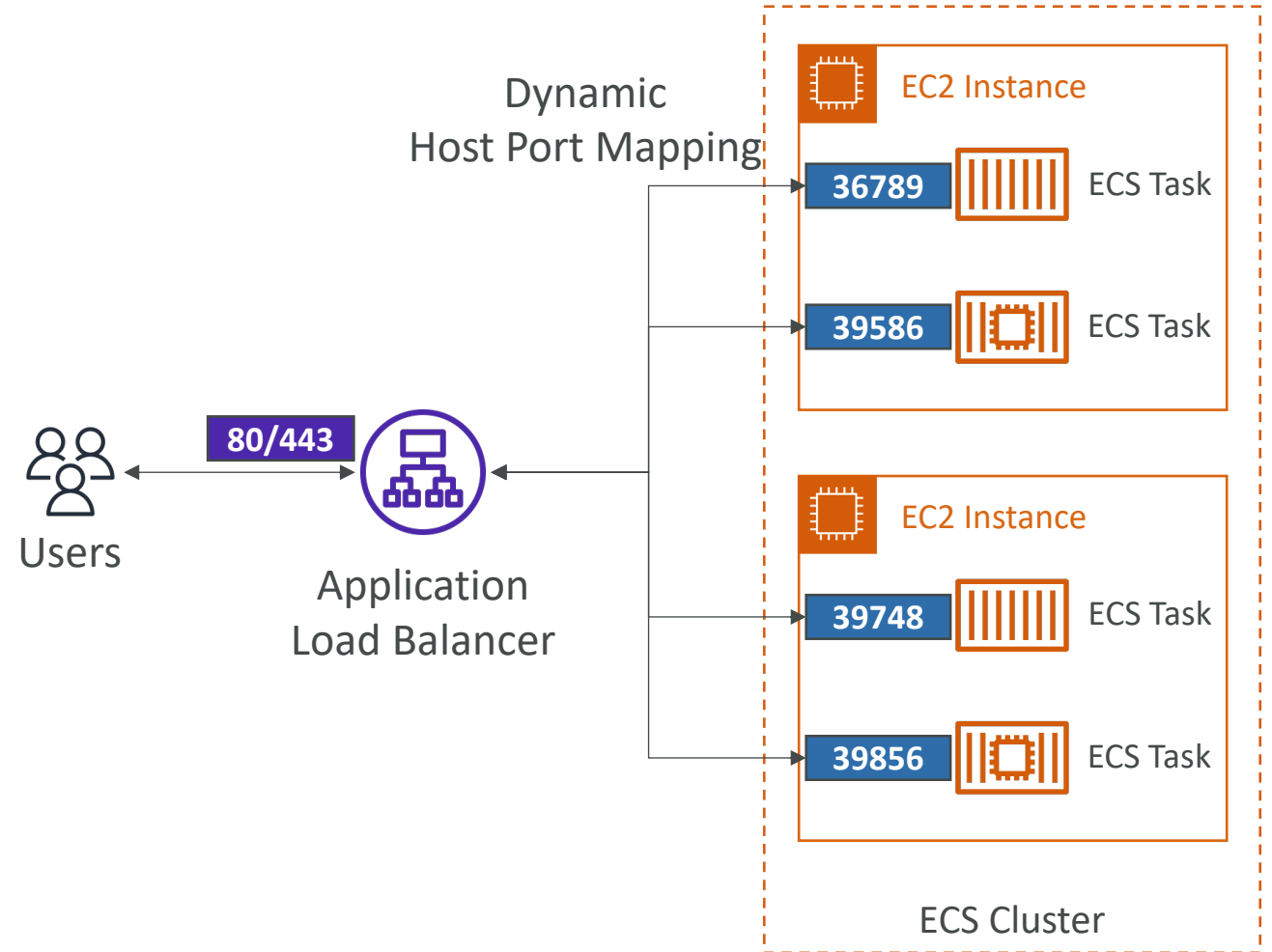
Amazon ECS – Task Definitions

- Task definitions are metadata in **JSON** form to tell ECS how to run a Docker container
- It contains crucial information, such as:
 - Image Name
 - Port Binding for Container and Host
 - Memory and CPU required
 - Environment variables
 - Networking information
 - IAM Role
 - Logging configuration (ex CloudWatch)
- Can define up to 10 containers in a Task Definition



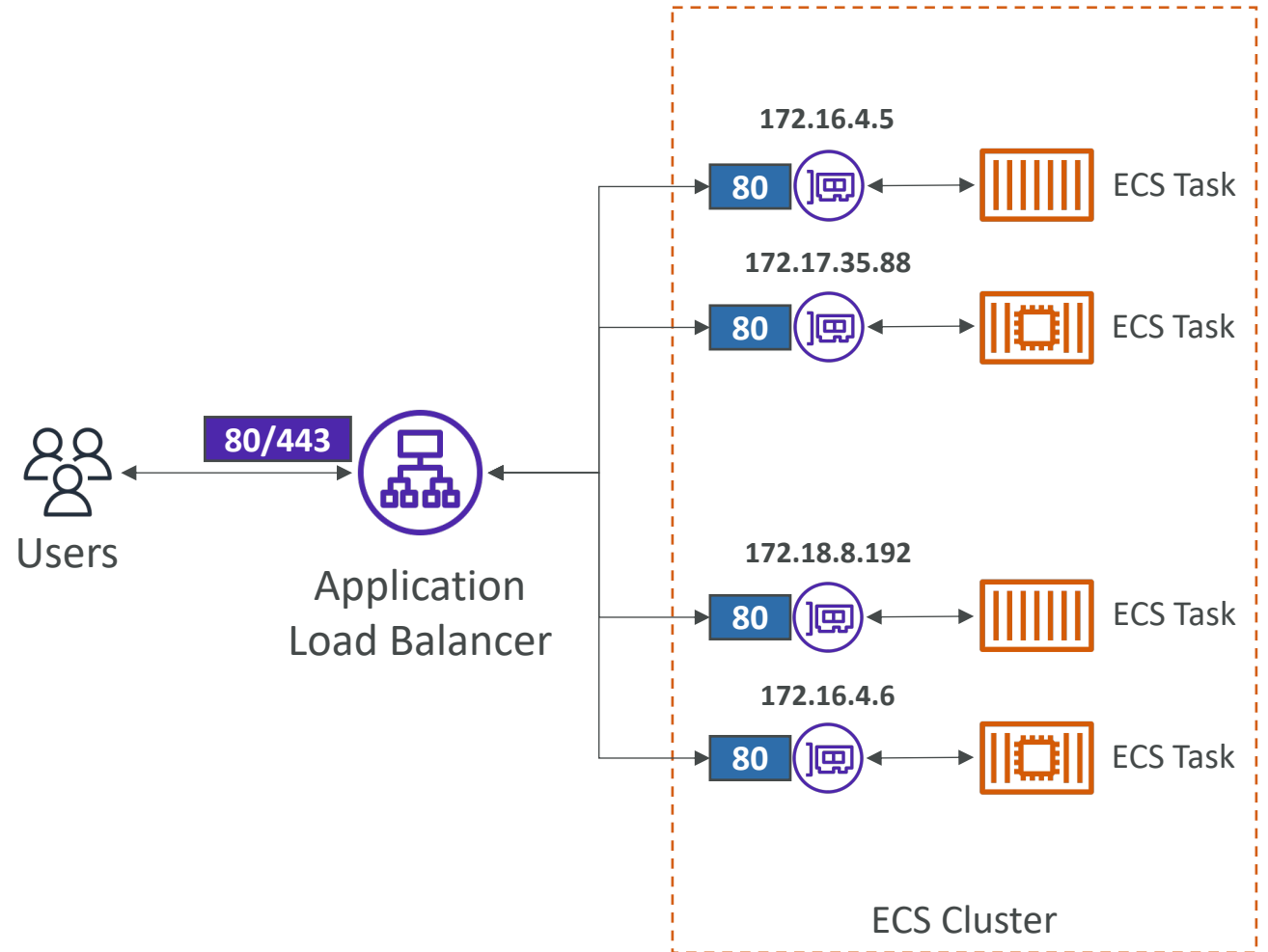
Amazon ECS – Load Balancing (EC2 Launch Type)

- We get a Dynamic Host Port Mapping if you define only the container port in the task definition
- The ALB finds the right port on your EC2 Instances
- You must allow on the EC2 instance's Security Group any port from the ALB's Security Group



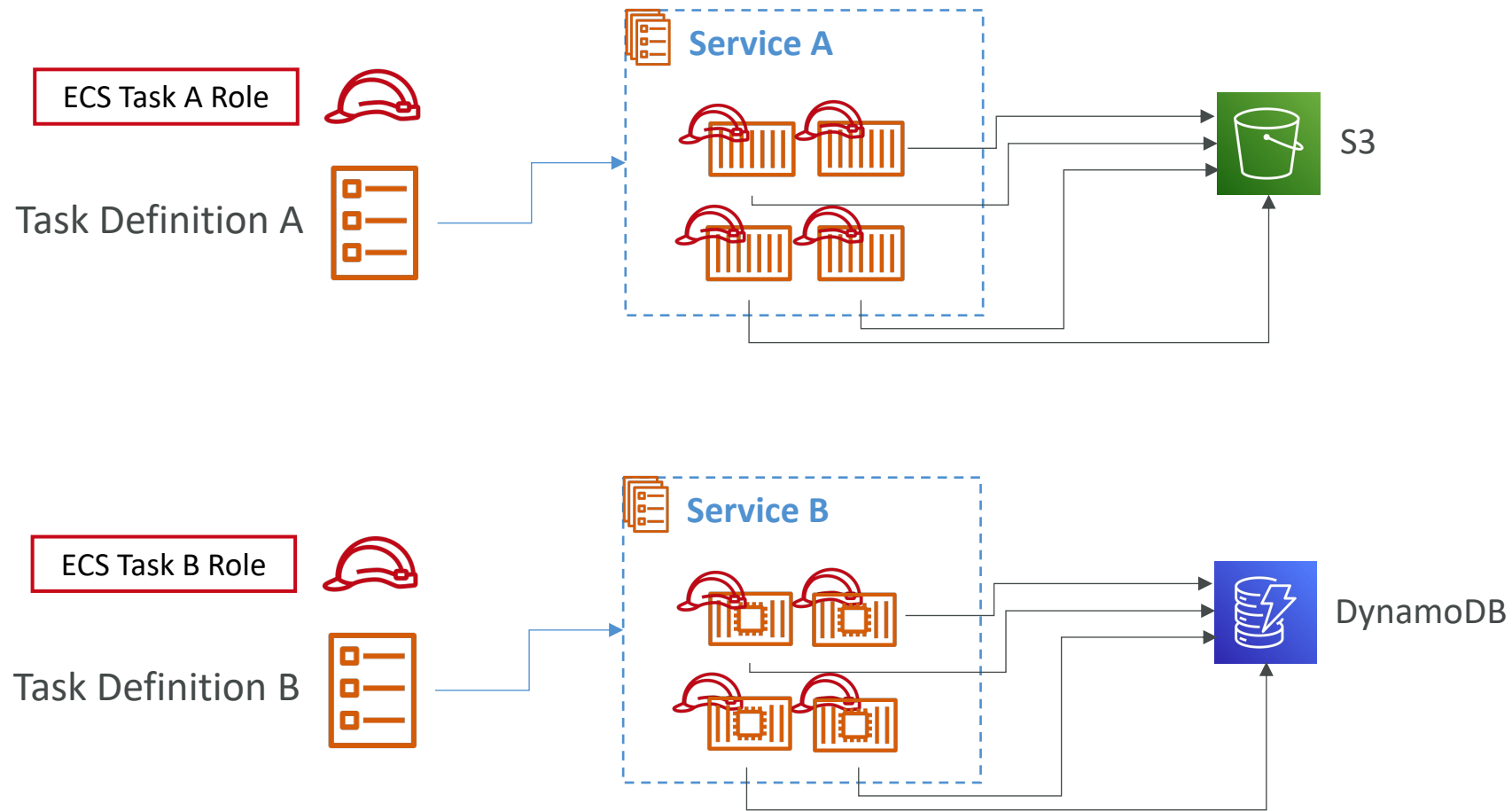
Amazon ECS – Load Balancing (Fargate)

- Each task has a **unique private IP**
- Only define the **container port** (host port is not applicable)
- Example
 - ECS ENI Security Group
 - Allow port 80 from the ALB
 - ALB Security Group
 - Allow port 80/443 from web



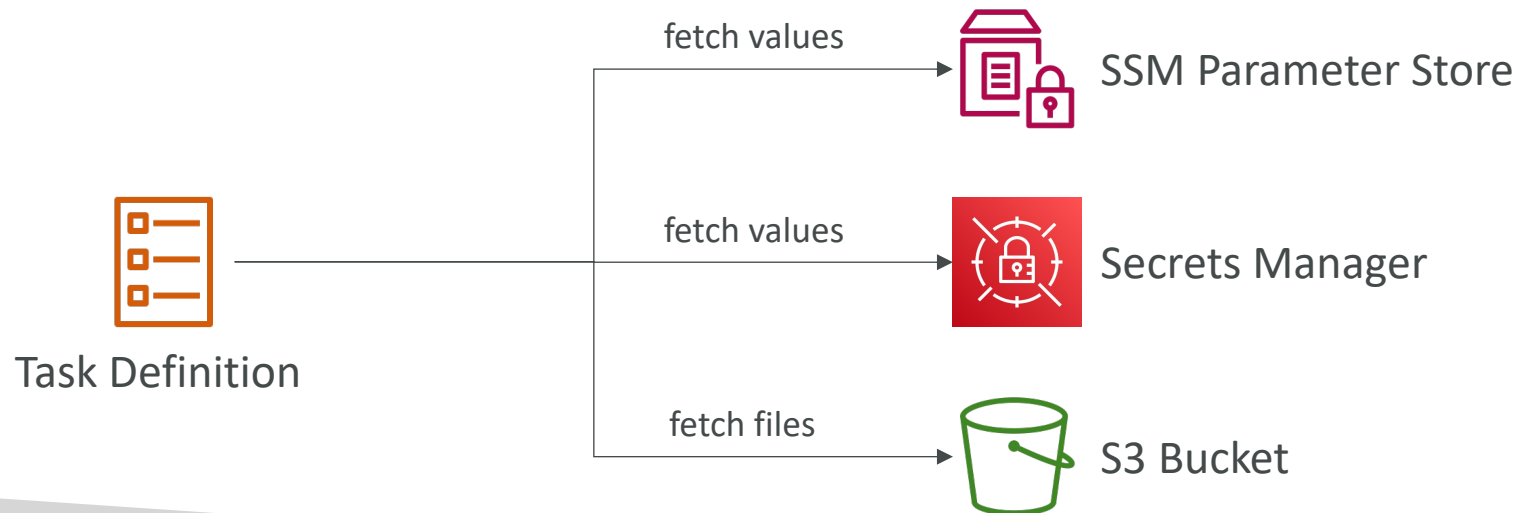
Amazon ECS

One IAM Role per Task Definition



Amazon ECS – Environment Variables

- Environment Variable
 - Hardcoded – e.g., URLs
 - SSM Parameter Store – sensitive variables (e.g., API keys, shared configs)
 - Secrets Manager – sensitive variables (e.g., DB passwords)
- Environment Files (bulk) – Amazon S3



Amazon ECS – Data Volumes (Bind Mounts)

- Share data between multiple containers in the same Task Definition
- Works for both **EC2** and **Fargate** tasks
- **EC2 Tasks** – using EC2 instance storage
 - Data are tied to the lifecycle of the EC2 instance
- **Fargate Tasks** – using ephemeral storage
 - Data are tied to the container(s) using them
 - 20 GiB – 200 GiB (default 20 GiB)
- Use cases:
 - Share ephemeral data between multiple containers
 - “Sidecar” container pattern, where the “sidecar” container used to send metrics/logs to other destinations (separation of concerns)

